

SANFORD, ME, USA

WMO: 726064

Lat: 43.394N

Lon: 70.708W

Elev: 244

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64709

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) 0.4	(d) -18.0	(e) 2.1	(f) 0.5	(g) -12.3	(h) 2.8	(i) 3.1	(j) 25.2	(k) 23.5	(l) 23.1	(m) 23.3	(n) 3.5	(o) 240	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	89.7	72.3	85.8	70.8	82.3	68.5	74.9	84.8	73.0	82.2	71.1	79.8	8.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	119.3	80.6	70.1	111.8	78.2	68.0	103.9	76.1	38.5	84.7	36.6	82.3	35.0	79.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.0	(b) 18.6	(c) 16.4	(e) -16.5	(f) 94.7	(g) 6.7	(h) 2.1	(i) -21.3	(j) 96.2	(k) -25.2	(l) 97.4	(m) -28.9	(n) 98.5	(o) -33.8	(p) 100.0		
(a) 21.0	(b) 18.6	(c) 16.4	(e) -16.8	(f) 77.8	(g) 6.0	(h) 1.7	(i) -21.2	(j) 79.1	(k) -24.7	(l) 80.1	(m) -28.1	(n) 81.1	(o) -32.5	(p) 82.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	21.5	23.7	32.2	43.8	53.7	63.5	69.3	66.9	59.5	48.3	37.8	28.0		
	DBStd	18.43	11.51	10.48	9.91	7.92	7.37	7.30	5.61	6.26	7.54	7.75	8.71	9.59		
	HDD50	3639	885	738	558	221	40	1	0	1	8	127	377	683		
	HDD65	7385	1349	1157	1016	638	360	114	21	50	198	519	816	1147		
	CDD50	2106	1	0	7	34	155	407	598	524	294	74	11	1		
	CDD65	379	0	0	0	1	11	70	154	109	33	1	0	0		
	CDH74	4245	0	0	14	51	242	794	1622	1150	343	29	0	0		
	CDH80	1313	0	0	4	14	78	255	542	327	91	2	0	0		
	WSAvg	6.1	7.0	7.3	7.6	7.2	6.1	5.2	4.6	4.4	4.7	6.0	6.3	6.6		
	PrecAvg	46.9	3.4	3.2	4.3	4.0	3.7	4.1	3.8	3.5	3.9	4.8	4.0	4.2		
(15) Precipitation	PrecMax	64.4	6.6	8.5	15.3	9.8	14.6	12.6	7.6	11.3	10.6	14.8	7.8	6.7		
	PrecMin	34.6	0.6	1.2	0.9	0.3	0.7	1.1	1.2	0.3	1.4	0.4	1.3	1.4		
	PrecStd	8.5	1.2	1.5	2.8	2.2	2.7	2.4	1.9	2.1	2.2	3.3	1.5	1.3		
	PrecStd	8.5	1.2	1.5	2.8	2.2	2.7	2.4	1.9	2.1	2.2	3.3	1.5	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	54.9	72.2	81.1	88.8	91.3	92.5	91.2	88.2	78.8	66.4	57.4		
		MCWB	52.2	48.5	56.2	60.7	66.9	73.3	74.7	74.5	72.8	63.0	57.3	50.5		
	2%	DB	47.7	48.1	58.8	71.7	81.3	87.6	89.7	87.5	82.0	72.1	61.4	51.6		
		MCWB	44.0	41.5	48.8	56.0	63.2	70.7	73.1	72.4	68.7	60.5	54.9	47.3		
	5%	DB	42.7	43.4	52.0	63.9	74.8	82.0	85.9	83.5	78.5	66.1	56.8	46.1		
		MCWB	37.9	37.1	43.5	51.4	59.5	67.7	71.3	70.2	66.4	58.1	50.3	41.9		
	10%	DB	37.0	38.7	46.1	59.1	69.9	78.9	82.2	81.0	73.2	62.8	53.4	42.6		
		MCWB	33.2	33.7	39.4	47.6	57.2	64.7	69.1	68.1	63.8	54.9	47.4	37.9		
	0.4%	WB	53.1	49.7	58.1	63.9	69.4	75.6	77.6	77.2	74.5	67.3	60.2	53.7		
		MCDB	55.0	53.3	70.6	75.5	83.4	87.0	88.8	87.7	83.9	71.6	64.1	56.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.7	42.8	50.2	58.5	65.4	72.8	75.2	74.2	71.5	63.0	56.3	47.1		
		MCDB	46.6	47.2	57.8	68.7	75.7	83.6	85.1	82.9	79.6	68.6	60.4	49.5		
	5%	WB	38.1	37.9	44.1	53.3	62.0	70.1	73.2	72.0	68.8	59.4	51.8	42.6		
		MCDB	41.4	42.2	49.7	61.8	72.4	78.9	82.7	80.3	74.8	65.5	56.0	46.5		
	10%	WB	33.6	34.6	40.0	48.8	58.7	67.3	71.2	70.1	66.1	56.0	47.6	38.0		
		MCDB	36.8	38.3	46.1	57.6	67.9	75.5	80.1	77.5	71.7	61.1	51.7	41.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	20.9	19.7	22.6	22.8	22.2	22.6	24.0	23.9	22.4	19.4	18.4		
		MCDBR	20.3	24.1	27.5	32.3	33.1	27.9	26.7	27.7	28.1	28.7	24.5	21.7		
		MCWBR	18.2	18.5	18.7	19.3	17.6	14.7	13.0	14.6	16.2	18.8	18.7	18.0		
		MCDBR	19.8	22.5	25.0	28.8	28.1	24.8	23.0	22.8	22.8	23.8	21.6	21.0		
	5% WB	MCWBR	18.3	18.5	18.3	19.2	16.1	14.3	12.5	13.1	14.5	17.6	18.2	19.0		
		MCWBR	18.3	18.5	18.3	19.2	16.1	14.3	12.5	13.1	14.5	17.6	18.2	19.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.296	0.320	0.352	0.388	0.437	0.454	0.422	0.373	0.331	0.311	0.290		
	taud		2.468	2.443	2.463	2.423	2.363	2.252	2.213	2.293	2.412	2.530	2.545	2.533		
	Ebn at Noon		272	288	293	290	281	266	259	265	271	272	260	260		
	Edh at Noon		23	28	31	35	38	43	44	39	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		540	815	1161	1461	1630	1750	1841	1675	1348	869	576	432		
	RadStd		36	62	72	157	185	148	122	84	78	79	52	47		

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+54

Nomenclature: See separate page